Practical 1

```
# create new collection

db.createCollection("students")


# insert one and many

db.students.insertOne({name:"Abhay", course: "Data Science", age: 21, grade: "A"})

db.students.insertMany([{name:"Akash", course: "Image Processing", age: 22, grade: "B"}, {name: "Harshika", course: "Networking", age: 21, grade: "A"}])


# find and conditional

db.students.find().pretty()

db.students.find({"age": { $gt: 21}})


# delete

db.students.deleteOne({ name: "Abhay" })


# update one and many

db.students.updateOne({name: "Harshika"}, { $set: {grade: "B"}})

db.students.updateMany({grade: "B"}, { $set: {subject: "Web Development"}})
```

Practical 2

```
# installation and starting
wsl --install
sudo apt update
sudo apt install redis-server
sudo service redis-server start
redis-cli ping  (it should be PONG)

redis-cli

# String
SET name "Alice"
GET name

# List
LPUSH fruits "Apple"
LPUSH fruits "Banana"
LPUSH fruits "Cherry"
LRANGE fruits 0 -1  --> -1 represents the last member, -2 last second member, 0 starting member

# Set
SADD colors "Red"
SADD colors "Blue"
SADD colors "Green"
SMEMBERS colors

# Expiry Example
SET temp "Hello" EX 10

# Increment / Decrement Example
INCR counter
DECR counter
```

Hash

HSET user:1000 name "Alice" age 25

HGETALL user:1000

#Sorted Set

ZADD scores 120 "abhay" 150 "harshika" 80 "akash"

ZRANGE scores 0 -1 WITHSCORES

Practical 3

how to start hbase

cd C:\Software\hbase-2.4.18\bin

start-hbase.cmd

jps (it will show HMaster, HRegionServer)

```
# start the shell to run code
hbase shell  (this will be shown - hbase(main):001:0>)

# create table
create 'employee','personal','job'
list   (it list all the tables)

# inserting
put 'employee','row1','personal:name','John'
put 'employee','row1','personal:age','30'
put 'employee','row1','job:role','Developer'
put 'employee','row1','job:salary','70000'

# read full table
scan 'employee'
# read specific table
get 'employee','row1'

# update
put 'employee','row1','job:role','Manager'
get 'employee','row1'   (use this to check and confirm chnages are done)

# delete
delete 'employee','row1','job:salary'  (deletes column value)
deleteall 'employee','row2'   (deletes entire row)

# disable and drop table
disable 'employee'
drop 'employee'

# remember to exit
exit
```

Practical 4

```
# start
cqlsh

# create namespace
CREATE KEYSPACE company WITH replication = {'class':'SimpleStrategy','replication_factor':1};
DESCRIBE KEYSPACES;

USE company;

# create table
CREATE TABLE employees ( employee_id UUID PRIMARY KEY, first_name TEXT, last_name TEXT, position TEXT,
salary DOUBLE, date_of_joining DATE );
DESCRIBE TABLES;

# insert values
INSERT INTO employees
(employee_id, first_name, last_name, position, salary, date_of_joining)
VALUES (uuid(), 'Abhay', 'Das', 'Software Engineer', 120000, '2022-06-01');

# reading values
SELECT * FROM employees;
SELECT first_name, position FROM employees;

# update values
UPDATE employees SET salary = 125000 WHERE employee_id = <employee_id>;

# delete values
DELETE FROM employees WHERE employee_id = <employee_id>;

# empty full table
```

TRUNCATE employees;

drop table

DROP TABLE employees;

Practical 5

#mongodb

```
db.students.find({age:{$gt:25}})
db.students.find({age:{$gt:25}}, {name:1, age:1})
db.students.find().sort({age:-1})
db.students.find().limit(2)
db.students.find().skip(1).limit(2)
```

hbase shell

list all tables

list

scan table

scan 'employee'

read specific data

get 'employee','row1'

filter data

scan 'employee', {FILTER => "SingleColumnValueFilter('personal','age',=,'binary:25')"}